

Migne Scanning System Operation

Manual

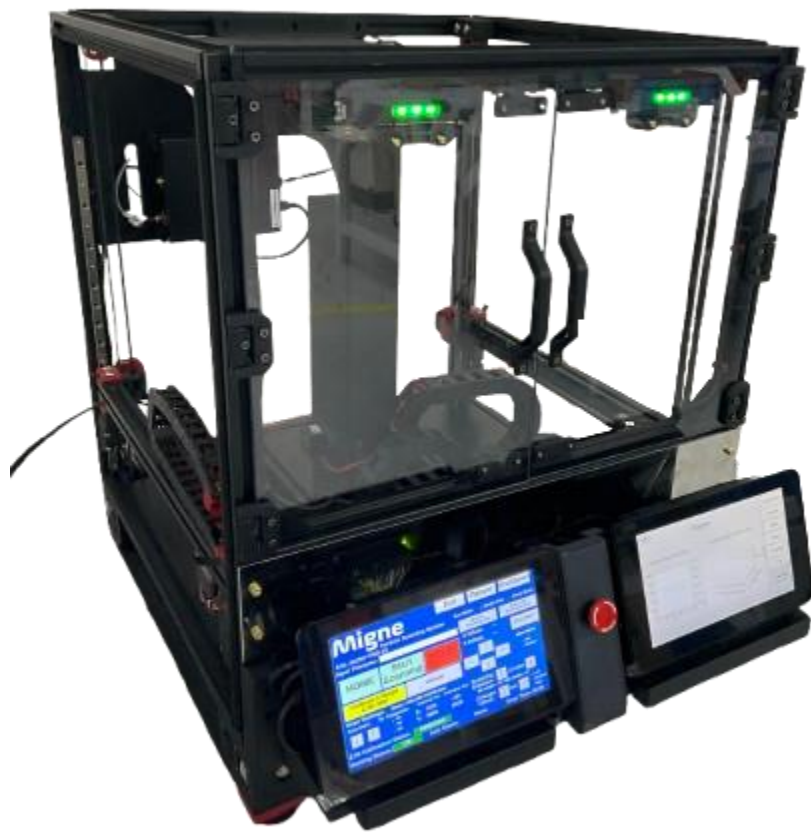


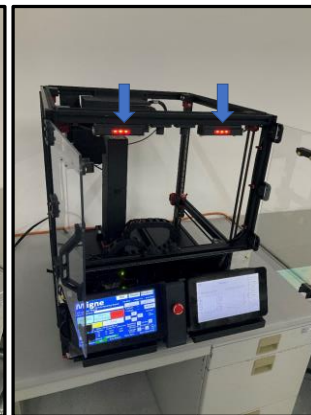
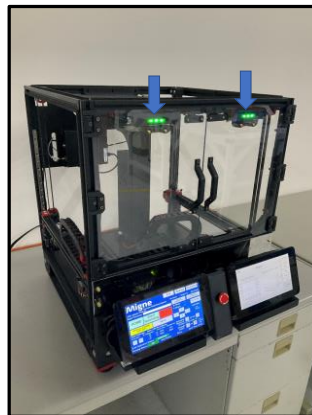
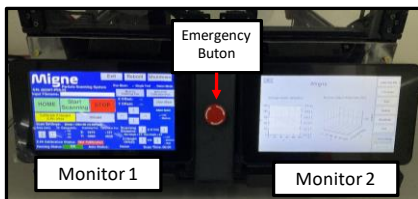
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1. Knowing the System

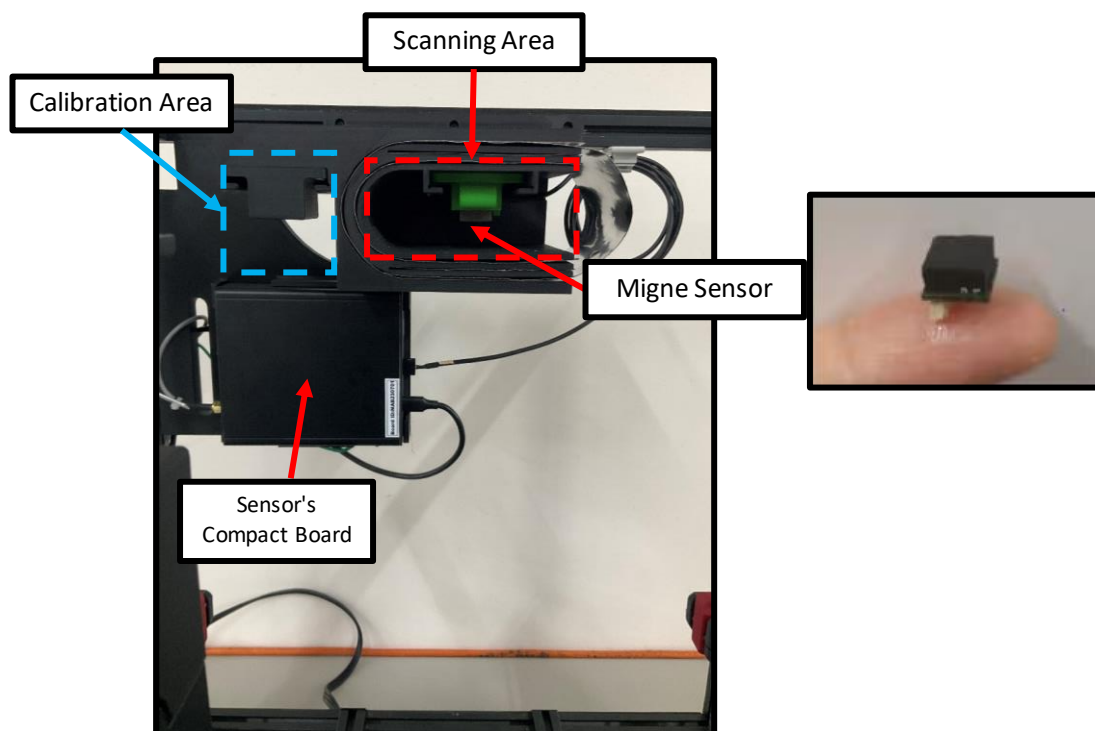
1.1 System Overview

- **Monitor 1** displays the main system controls and operation interface.
- **Monitor 2** displays the live scanning plot for foreign particle detection.
- The **Emergency Button** immediately stops all mechanical operations of the system.
- The **Safety Door Sensor** functions the same as the Emergency Button; when the door is open, the system will stop automatically.
- **Door Status Indicator:** Green: Door Closed, Red: Door Open.



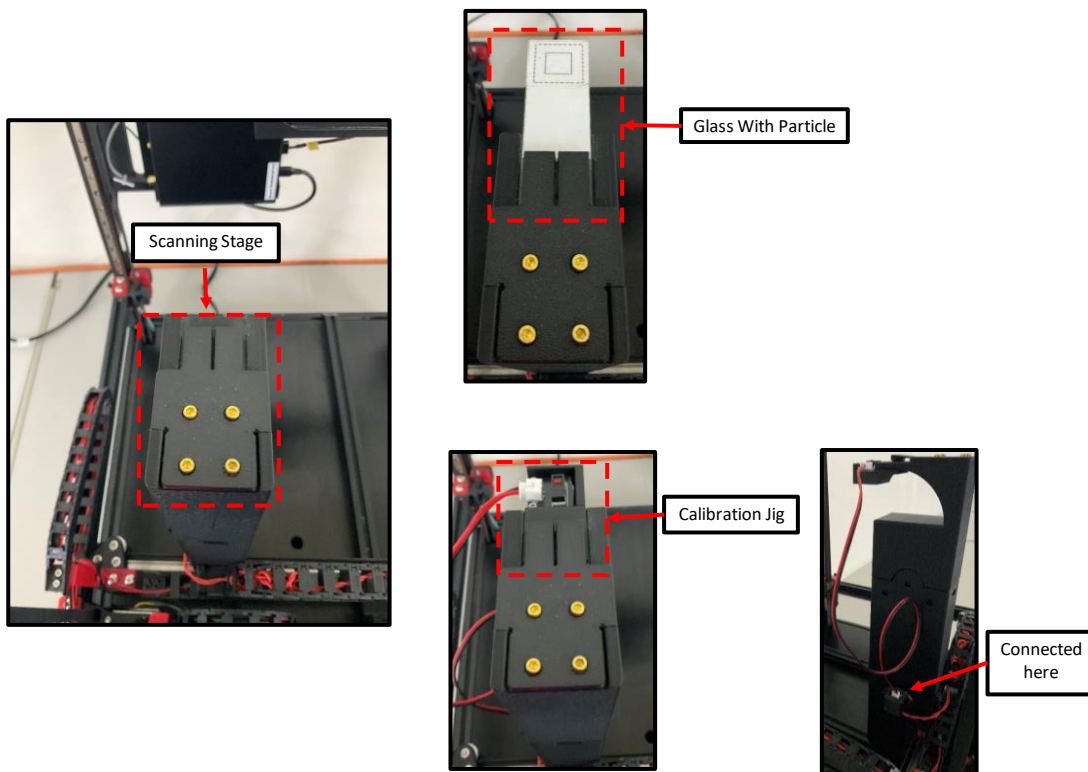
1.2 Calibration & Scanning Areas

- The **Calibration Area** is used to calibrate the Z-Height of the scanning stage. This ensures that the distance between the sensor and the surface is set to the default **2.0 mm**.
- The **Scanning Area** is where foreign particle detection is performed.



1.3 Scanning Stage

- The **Scanning Stage** is responsible for the movement during scanning detection.
- This is where the **glass with particle** is placed during scanning.
- The **Calibration Jig** is also mounted here when performing Z-Height calibration.

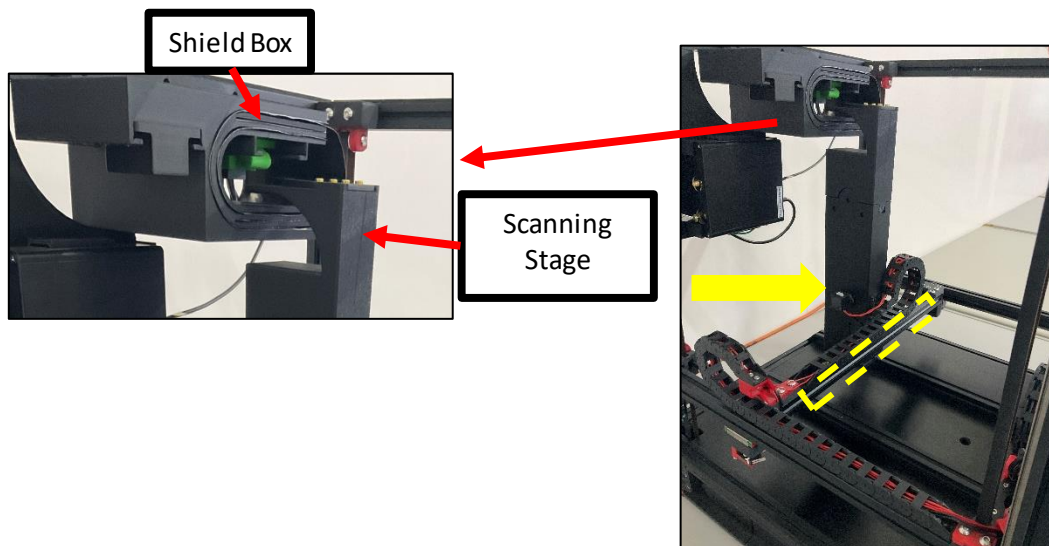


2. Startup Procedure

2.1 Stage Position Check

Ensure that the scanning stage is **not inside the Shield Box**.

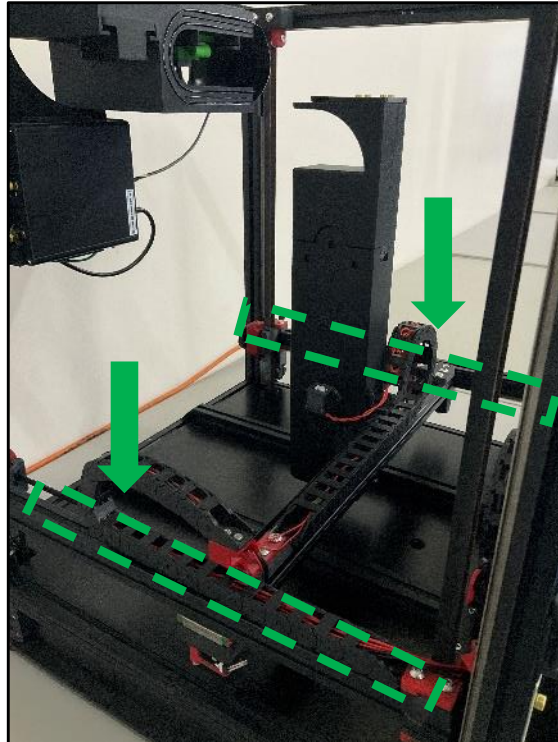
If the stage is inside, pull the **Y-Axis Frame** until the stage is completely outside the Shield Box.



2.2 Y-Axis Frame Adjustment

Push down the Y-Axis Frame until it reaches the **lowest position**.

Make sure that all sides are **level and even**.



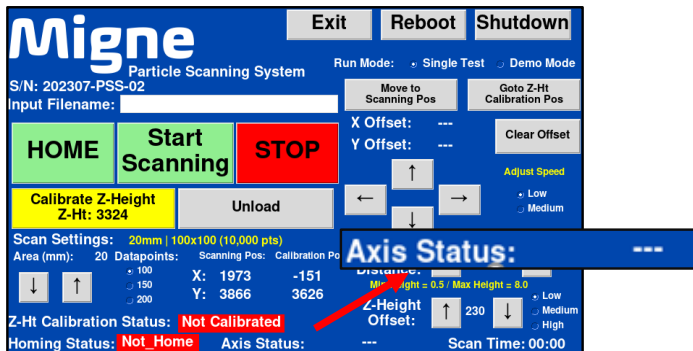
2.3 Power On

Turn **ON** the Power Switch and wait for the system to complete the boot-up process.

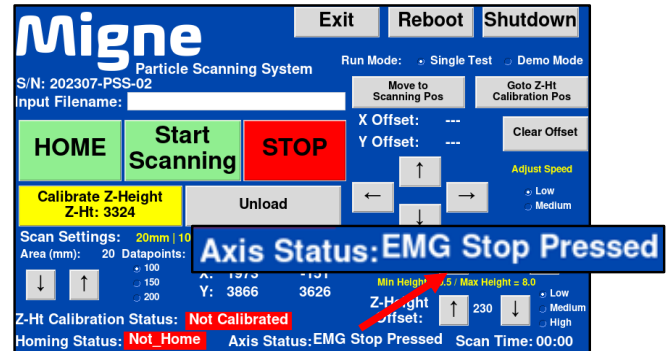
2.4 Emergency Button Check

Ensure that the **Emergency Button** is released before powering on.

The system will not operate if the Emergency Button is pressed until it is released



Interface when the EMG button is not pressed



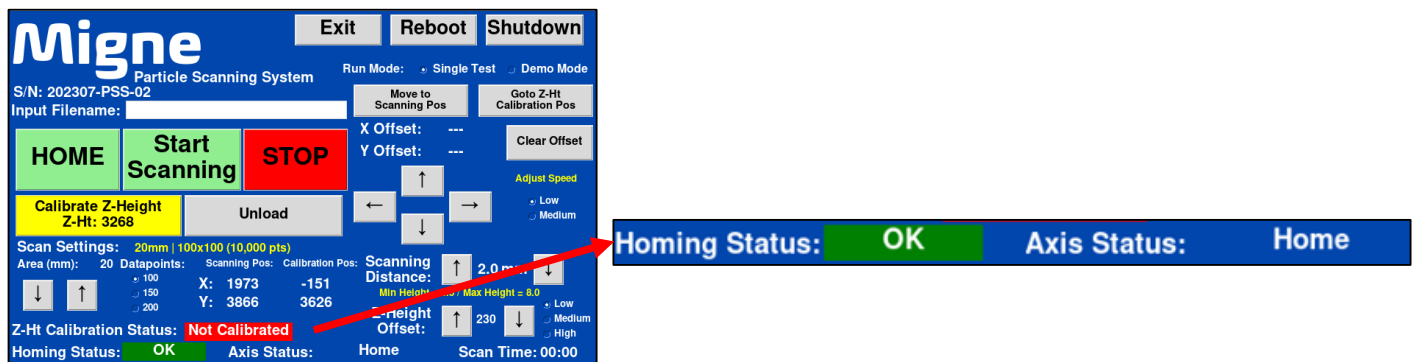
Interface when the EMG button is pressed

2.5 Homing

On the Monitor 1, press the HOME button

2.6 Homing Confirmation

Once finished homing, Homing Status will display "OK" and Axis Status will display "Home".



3. Setup Procedure

3.1 Z-Height Calibration Position Offset

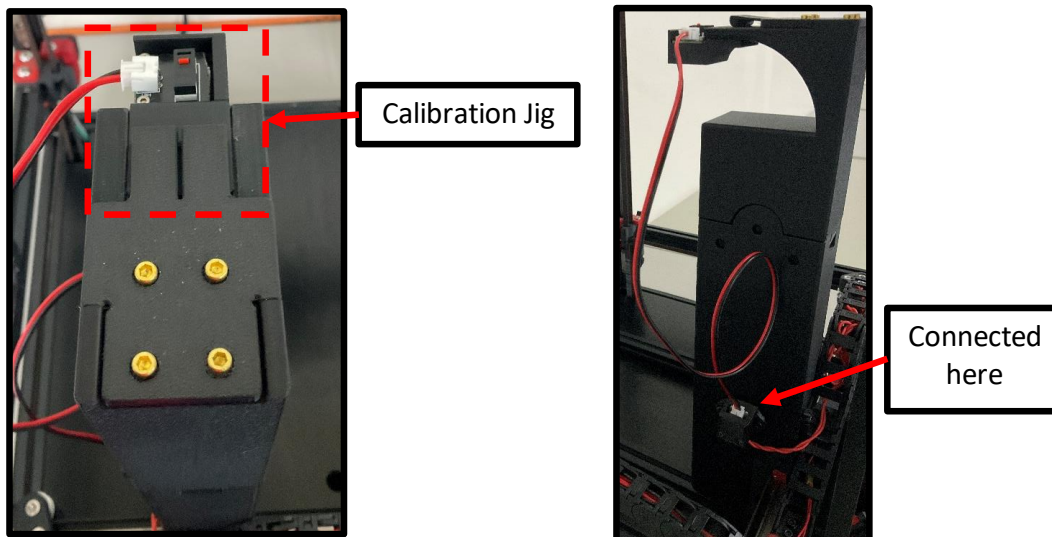
Press **Go to Z-Ht Calibration Pos.**

If no offset is indicated, set the offset using the arrow keys.

3.2 Calibration Jig Setup

After setting the Calibration Position offset:

1. Press the **HOME** button.
2. Place the **Calibration Jig** onto the scanning stage.
3. Connect the Calibration Jig properly.



3.3 Z-Height Calibration

Press the **Calibration** button to calibrate the Z-Height of the scanning system.

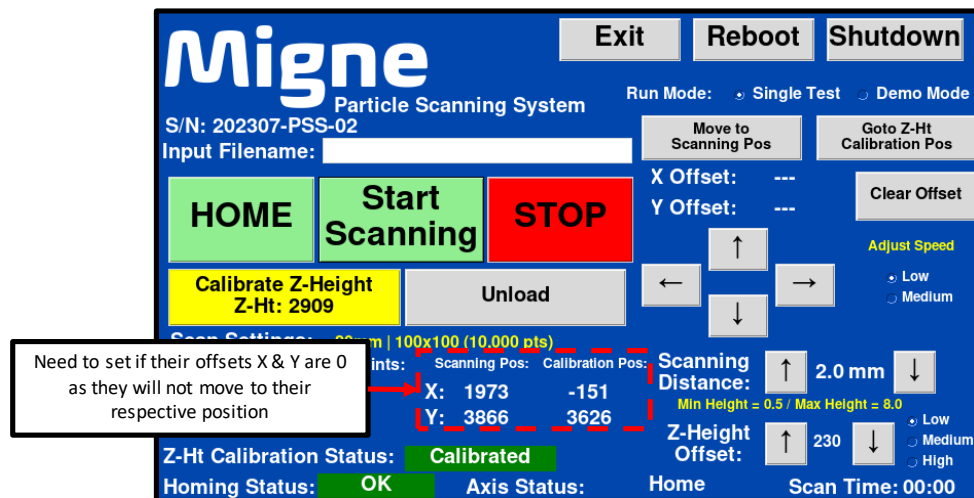
3.4 Scanning Position Offset

After calibration:

- If the **Scanning Position offset** is not set, press Move to Scanning Pos.
- Adjust the offset using the arrow keys.
- Once set, perform Home positioning.

Note:

If X and Y offsets are set to 0, the stage will not move to their respective positions.



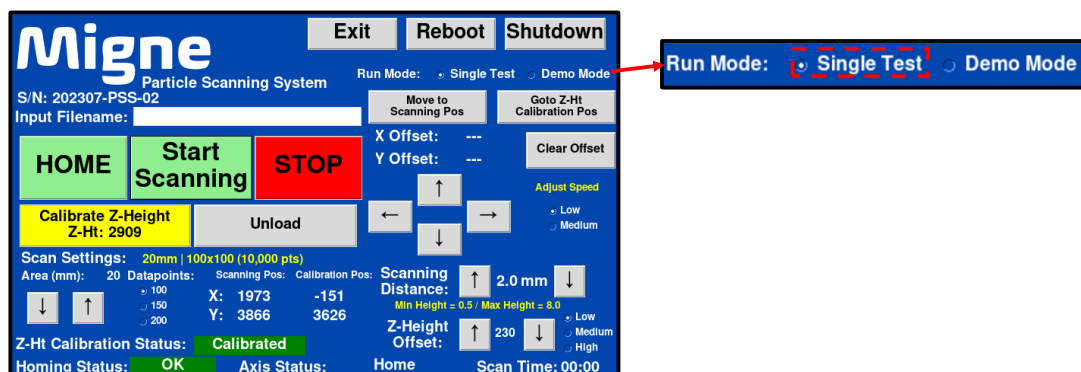
4. Scanning Procedure

4.1 Homing Check

Perform **Home positioning** if the Home Status is not currently set to **HOME**.

4.2 Test Mode Selection

Ensure that the **Single Test Option** is selected.



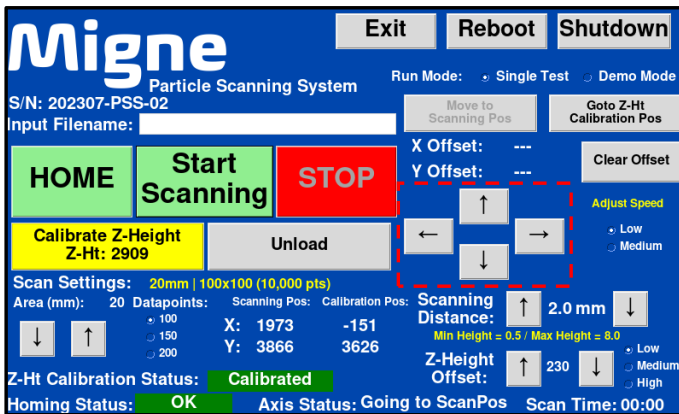
4.3 Sample Placement

Place the **glass with particle** onto the scanning stage, then press **Move to Scanning Pos.**

4.4 Sample Alignment

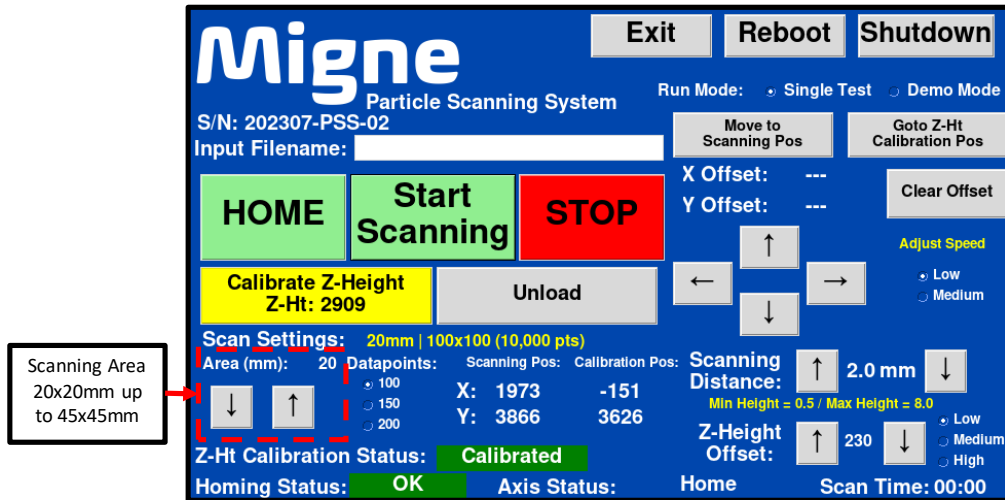
Using the arrow keys:

- Center the glass and particle under the sensor.
- After adjustment, press the **HOME** button.



4.5 Scanning Parameters

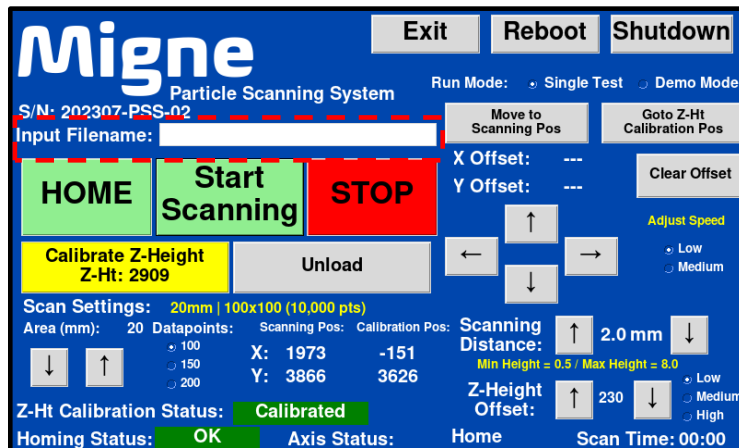
- Select the desired **scanning distance**.
- Choose the **scanning area** from 20 × 20 mm up to 45 × 45 mm.



4.5 Start Scanning

Enter the desired filename in the **Filename Input Box**.

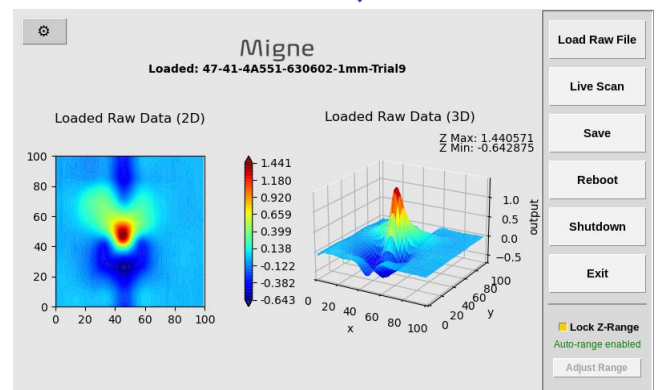
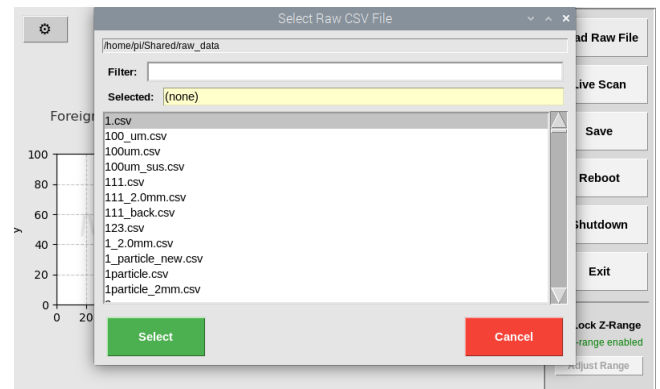
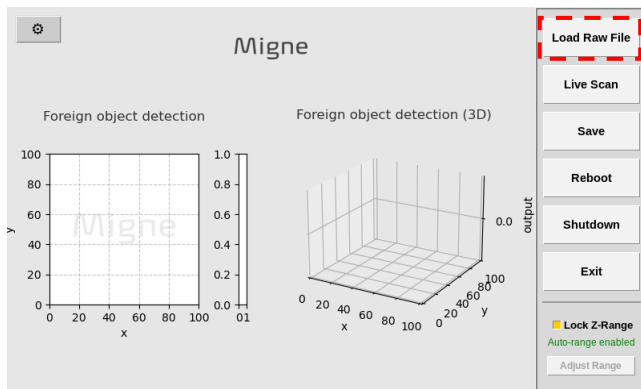
After confirming all parameters, press **Start Scanning**.



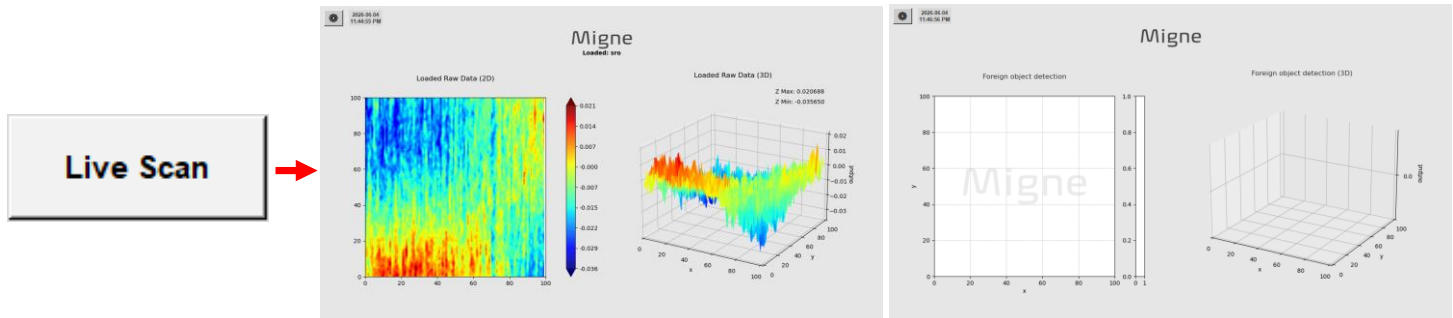
5. Getting the Scan Data Results

5.1 Displaying the Scan Detection Results:

- On Monitor 2, Press **Load Raw File**
- Input the filename you want to display on the Monitor 2
- After selecting, it will display the scan detection of the raw file you selected.



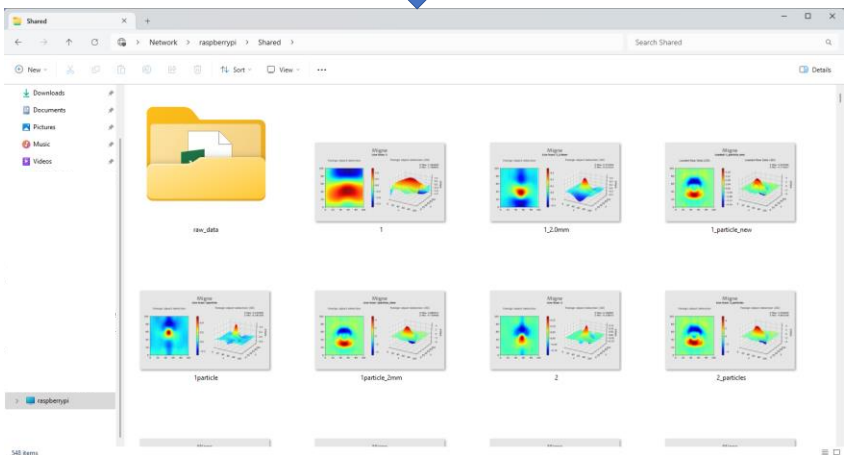
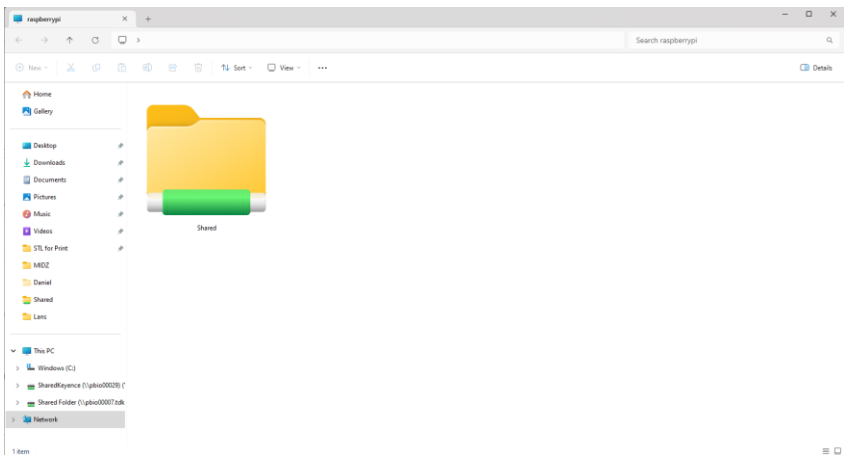
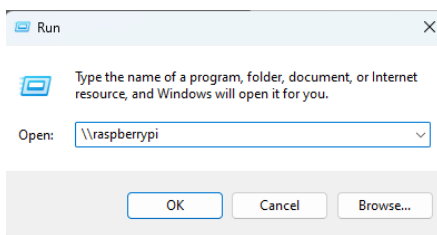
- In **Live Scan**, it will reset back the display from the loaded data



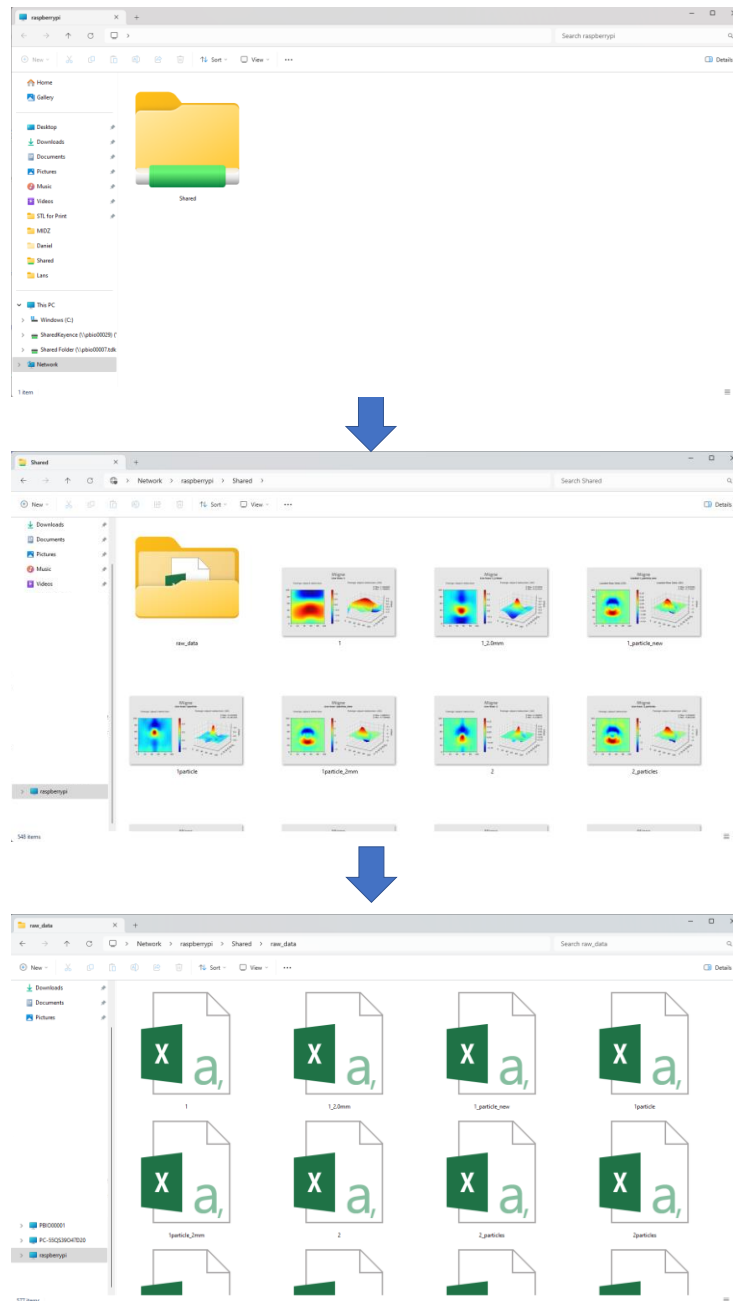
5.2 Downloading the Measurement Raw Data and Images

Results:

- Connect a **LAN cable** between:
 1. Monitor 2 Ethernet port
 2. PC/Laptop Ethernet port
- On the PC, open **Run** and type:
- \\raspberrypi
- Open the Shared folder for Scan Images

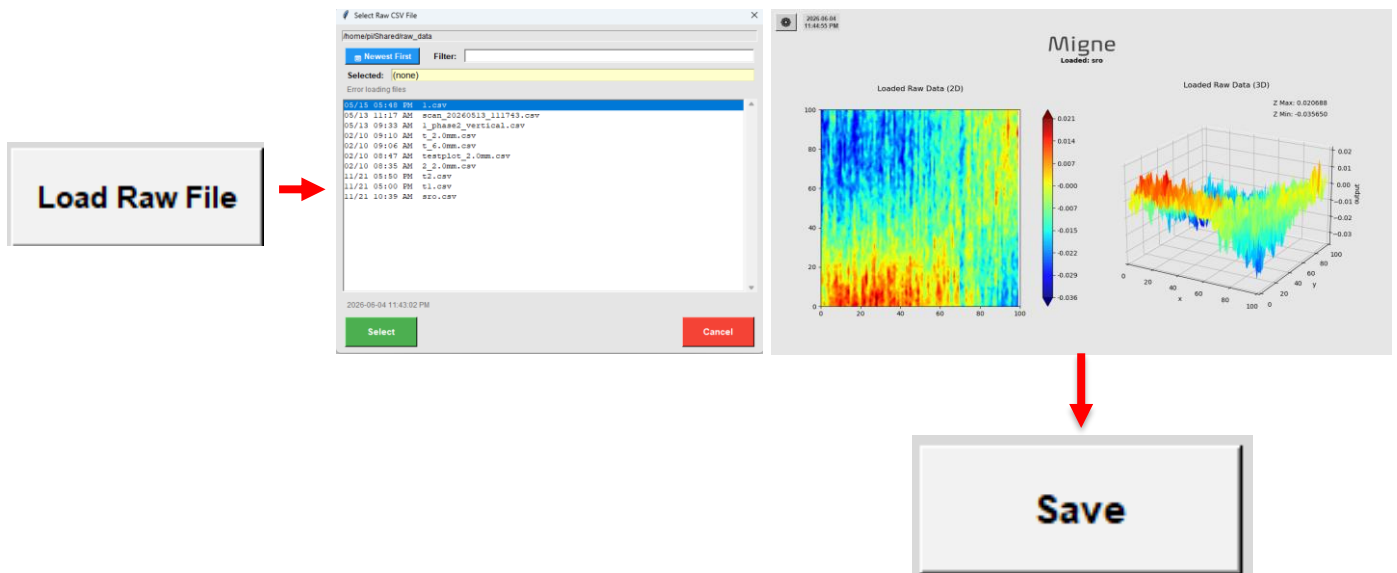


- Open the Shared/raw_data folder for raw data (Excel File).



- Copy and Paste it in your Pc/Laptop storage.

- Using the **USB port**, mount the **flash drive**
- Load the raw data you want to transfer, then click **Save**.
- It will directly transfer the raw data and its image display to the USB.



- After transferring, press **Eject USB** to unmount properly.

